frame

Literally, a packet as it is transmitted across a serial line. The term derives from character oriented protocols that added special start-of-frame and end-of-frame characters when transmitting packets. We use the term throughout this book to refer to the objects that physical networks transmit.

Frame Relay

The name of a connection-oriented network technology that is offered by telephone companies.

FTP

(*File Transfer Protocol*) The TCP/IP standard, high-level protocol for transferring files from one machine to another. FTP uses TCP.

full duplex

Characteristic of a technology that allows simultaneous transfer of data in two directions. TCP provides full duplex connections.

FYI

(*For Your Information*) A subset of the RFCs that contain tutorials or general information about topics related to TCP/IP or the connected Internet.

gated

(*GATEway Daemon*) A program run on a router that uses an IGP to collect routing information from within one autonomous system and EGP to advertise the information to another autonomous system.

gateway

Any mechanism that connects two or more heterogeneous systems and translates among them. Originally, researchers used the term *IP gateway* for dedicated computers that route IP datagrams; vendors have adopted the term *IP router*.

gateway requirements

See *router requirements*.

Gbps

(*Giga Bits Per Second*) A measure of the rate of data transmission equal to 2^{30} bits per second. Also see *Kbps*, *Mbps*, and *baud*.

GGP

(*Gateway to Gateway Protocol*) The protocol originally used by core gateways to exchange routing information. GGP is now obsolete.

gopher

An early menu-driven information service used in the Internet.

GOSIP

(*Government Open Systems Interconnection Profile*) A U.S. government procurement document that specified agencies may use OSI protocols in new networks after August 1991. Although GOSIP was originally thought to eliminate the use of TCP/IP on government internets, clarifications have specified that government agencies can continue to use TCP/IP.

graceful shutdown
A protocol mechanism that allows two communicating parties to agree to terminate communication without confusion even if underlying packets are lost, delayed, or duplicated. TCP uses a 3-way handshake to guarantee graceful termination.

graft
An operation in which a multicast router joins a shared forwarding tree; the opposite of *prune*.

GRE
(*Generic Routing Encapsulation*) A scheme for encapsulating information in IP that includes IP-in-IP as one possibility.

H.323
An ITU recommendation for a suite of protocols used for IP telephony.

half duplex
Characteristic of a technology that only permits data transmission in one direction at a time. Compare to *full duplex*.

hardware address
The low-level addresses used by physical networks. Synonyms include *physical address* and *MAC address*. Each type of network hardware has its own addressing scheme (e.g., an Ethernet address is 48 bits).

header
Information at the beginning of a packet or message that describes the contents and specifies a destination.

HELLO
A protocol used on the original NSFNET backbone. Although obsolete, Hello is interesting because it uses delay as the routing metric and chooses a path with minimum delay.

HELO
The command on the initial exchange of the SMTP protocol.

hierarchical addressing
An addressing scheme in which an address can be subdivided into parts that each identify successively finer granularity. IP addresses use a two-level hierarchy in which the first part of the address identifies a network and the second part identifies a particular host on that network. Routers use the network portion to forward a datagram until the datagram reaches a router that can deliver it directly. Subnetting introduces additional levels of hierarchical routing.

historic
An IETF classification used to discourage the use of a protocol. In essence, a program that is declared historic is obsolete.

hold down
A short fixed time period following a change to a routing table during which no further changes are accepted. Hold down helps avoid routing loops.

hop count

A measure of distance between two points in an internet. A hop count of *n* means that *n* routers separate the source and destination.

hop limit

The IPv6 name for the datagram header field that IPv4 calls *time to live*. The hop limit, which prevents datagrams from following a routing loop forever, is decremented by each router along the path.

host

Any end-user computer system that connects to a network. Hosts include devices such as printers, small notebook computers, as well as large supercomputers. Compare to *router*.

host requirements

A long document that contains revisions and updates of many TCP/IP protocols. The host requirements document is published in a pair of RFCs. See *router requirements*.

host-specific route

An entry in a routing table that refers to a single host computer as opposed to routes that refer to a network, an IP subnet, or a default.

HTML

(*HyperText Markup Language*) The standard document format used for Web pages.

HTTP

(*Hypertext Transfer Protocol*) The protocol used to transfer Web documents from a server to a browser.

hub

An inexpensive electronic device to which multiple computers attach, usually using twisted pair wiring, to send and receive packets. A hub operates at layer 2 by replicating signals. Ethernet hubs are especially popular.

IAB

(*Internet Architecture Board*) A small group of people who set policy and direction for TCP/IP and the global Internet. The IAB was formerly known as the *Internet Activities Board*. See *IETF*.

IAC

(*Interpret As Command*) An escape used by TELNET to distinguish commands from normal data.

IANA

(*Internet Assigned Number Authority*) Essentially one individual (Jon Postel), IANA was originally responsible for assigning IP addresses and the constants used in TCP/IP protocols. Replaced by ICANN in 1999.

ICANN

(*Internet Corporation For Assigned Names and Numbers*) The organization that took over the IANA duties after Postel's death.

ICCB

(*Internet Control and Configuration Board*) A predecessor to the IAB.

ICMP

(*Internet Control Message Protocol*) An integral part of the Internet Protocol (IP) that handles error and control messages. Specifically, routers and hosts use ICMP to send reports of problems about datagrams back to the original source that sent the datagram. ICMP also includes an echo request/reply used to test whether a destination is reachable and responding.

ICMPv6

(*Internet Control Message Protocol version 6*) The version of ICMP that has been defined for use with IPv6.

IEN

(*Internet Engineering Notes*) A series of notes developed in parallel to RFCs. Although the series is obsolete, some IENs contain early discussion of TCP/IP and the Internet not found in RFCs.

IESG

(*Internet Engineering Steering Group*) A committee consisting of the IETF chairperson and the area managers. The IESG coordinates activities among the IETF working groups.

IETF

(*Internet Engineering Task Force*) A group of people under the IAB who work on the design and engineering of TCP/IP and the global Internet. The IETF is divided into areas, which each has an independent manager. Areas are further divided into working groups.

IGMP

(*Internet Group Management Protocol*) A protocol that hosts use to keep local routers apprised of their membership in multicast groups. When all hosts leave a group, routers no longer forward datagrams that arrive for the group.

IGP

(*Interior Gateway Protocol*) The generic term applied to any protocol used to propagate network reachability and routing information within an autonomous system. Although there is no single standard IGP, RIP is among the most popular.

IMP

(*Interface Message Processor*) The original term for packet switches in the ARPANET; now loosely applied to a switch in any packet network.

InATMARP

(*Inverse ATM ARP*) Part of the address resolution protocol needed for non-broadcast multiple access networks such as ATM.

indirect delivery

Delivery of a datagram through a router as opposed to a direct transmission from the source host to the destination host.

INOC

(*Internet Network Operations Center*) Originally, a group of people at BBN that monitored and controlled the Internet core gateway system. Now applied to any group that monitors an internet.

inter-autonomous system routing

Also known as exterior routing. BGP-4 is currently the most popular protocol for exterior routing.

International Organization for Standardization

See *ISO*.

International Telecommunications Union (ITU)

An international organization that sets standards for interconnection of telephone equipment. It defined the standards for X.25 network protocols. (Note: in Europe, *PTTs* offer both voice telephone services and X.25 network services).

internet

Physically, a collection of packet switching networks interconnected by routers along with TCP/IP protocols that allow them to function logically as a single, large, virtual network. When written in upper case, Internet refers specifically to the global Internet.

Internet

The collection of networks and routers that spans over 200 countries, and uses TCP/IP protocols to form a single, cooperative virtual network.

Internet address

See *IP address*.

Internet Draft

A draft document generated by the IETF; if approved, the document will become an RFC.

Internet Protocol

See IP.

Internet Society

The non-profit organization established to foster interest in the Internet. The Internet Society is the host organization of the IAB.

Internet worm

A program designed to travel across the Internet and replicate itself endlessly. When a student released the Internet worm, it made the Internet and many attached computers useless for hours.

interoperability

The ability of software and hardware on multiple machines from multiple vendors to communicate meaningfully. This term best describes the goal of internetworking, namely, to define an abstract, hardware independent networking environment that makes it possible to build distributed computations that interact at the network transport level without knowing the details of underlying technologies.

Intranet

A private corporate network consisting of hosts, routers, and networks that use TCP/IP technology. An intranet may or may not connect to the global Internet.

IP

(*Internet Protocol*) The TCP/IP standard protocol that defines the IP datagram as the unit of information passed across an internet and provides the basis for connectionless, best-effort packet delivery service. IP includes the ICMP control and error message protocol as an integral part. The entire protocol suite is often referred to as TCP/IP because TCP and IP are the two fundamental protocols.

IP address

A 32-bit address assigned to each host that participates in a TCP/IP internet. IP addresses are the abstraction of physical hardware addresses just as an internet is an abstraction of physical networks. To make routing efficient, each IP address is divided into a network portion and a host portion.

IP datagram

The basic unit of information passed across a TCP/IP internet. An IP datagram is to an internet as a hardware packet is to a physical network — each datagram contains a source and destination address along with data.

IP gateway

A synonym for *IP router.*

IP-in-IP

The encapsulation of one IP datagram inside another for transmission through a tunnel. IP in IP is often used to send multicast datagrams across the Internet.

IP multicast

An addressing and forwarding scheme that allows transmission of IP datagrams to a subset of hosts. The Internet currently does not have extensive facilities for routing IP multicast.

IP router

A device that connects two or more (possibly heterogeneous) networks and passes IP traffic between them. As the name implies, a router looks up the datagram's destination address in a routing table to choose a next hop.

IP switching

Originally a high-speed IP forwarding technology developed by Ipsilon Corporation, now generally used in reference to any of several similar technologies.

IP telephony

A telephone system that uses IP to transport digitized voice.

IPng

(*Internet Protocol — the Next Generation*) A term applied to all the activities surrounding the specification and standardization of the next version of IP. Also see *IPv6.*

IPsec

(*IP SECurity*) A security standard that allows the sender to choose to authenticate or encrypt a datagram. IPsec can be used with either IPv4 or IPv6.

IPv4

(*Internet Protocol version 4*). The official name of the current version of IP.

IPv6

(*Internet Protocol version 6*). The name of the next version of IP. Also see *IPng.*

IRSG

(*Internet Research Steering Group*) The group of people who head the IRTF.

IRTF

(*Internet Research Task Force*) A group of people working on research problems related to TCP/IP and the connected Internet. The IRTF is not as active as the IETF.

ISDN

(*Integrated Services Digital Network*) The name of the digital network service that telephone carriers provide.

ISO

(*International Organization for Standardization*) An international body that drafts, discusses, proposes, and specifies standards for network protocols. ISO is best know for its 7-layer reference model that describes the conceptual organization of protocols. Although it has proposed a suite of protocols for Open System Interconnection, the OSI protocols have not been widely accepted in the commercial market.

ISOC

Abbreviation for *Internet SOCiety.*

isochronous

Characteristic of a network system that does not introduce jitter. The conventional telephone system is isochronous.

ISODE

(*ISO Development Environment*) Software that provides an ISO transport level protocol interface on top of TCP/IP. ISODE was designed to allow researchers to experiment with ISO's higher-level OSI protocols without requiring an internet that supports the lower levels of the OSI suite.

ISP

(*Internet Service Provider*) Any organization that sells Internet access, either permanent connectivity or dialup access.

ITU

Abbreviation for the *International Telecommunication Union*, a standards organization.

jitter

A technical term used to describe unwanted variance in delay caused when one packet in a sequence must be delayed more than another. The typical cause of jitter is other traffic on a network.

Karn's Algorithm

An algorithm that allows transport protocols to distinguish between valid and invalid round-trip time samples, and thus improve round-trip estimations.

Kbps

(*Kilo Bits Per Second*) A measure of the rate of data transmission equal to 2^{10} bits per second. Also see *Gbps*, *Mbps*, and *baud*.

keepalive

A small message sent periodically between two communicating entities to ensure that network connectivity remains intact and that both sides are still responding. BGP uses keepalives.

LAN

(*Local Area Network*) Any physical network technology designed to span short distances (up to a few thousand meters). Usually, LANs operate at tens of megabits per second through several gigabits per second. Examples include Ethernet and FDDI. See *MAN* and *WAN*.

layer 1

A reference to the hardware interface layer of communication. The name is derived from the ISO 7-layer reference model. Layer 1 specifications refer to physical connections, including connector configuration and voltages on wires. (Sometimes called *level 1*.)

layer 2

In the ISO 7-layer model, a reference to link level communication (e.g., frame format). In the TCP/IP 5-layer model, layer 2 refers to physical frame format and addressing. Thus, a layer 2 address is a MAC address (e.g., an Ethernet address).

layer 3

In the ISO 7-layer model, a reference to the network layer. In the TCP/IP 5-layer model, a reference to the internet layer (IP and the IP datagram format). Thus, an IP address is a layer 3 address.

leaf

A graph-theoretic term for a router or a network at the "edge" of an internet.

link-local address

An address used with IPv6 that has significance only on a single network.

link state routing

One of two approaches used by routing protocols in which routers broadcast status messages and use Dijkstra's SPF algorithm to compute shortest paths. See *distance vector routing*.

link status routing

A synonym for *link state routing*.

LIS
(*Logical IP Subnet*) A group of computers connected via ATM that use ATM as an isolated local network. A computer in one LIS cannot send a datagram directly to a computer in another LIS.

little endian
A format for storage or transmission of binary data in which the least-significant byte (bit) comes first. See *big endian.*

LLC
(*Logical Link Control*) One of the fields in an NSAP header.

logical subnet
An abbreviation of *Logical IP Subnet* (*LIS*).

long haul network
Older term for *wide area network* (*WAN*).

longest-prefix matching
The technique used by IP routers when searching a routing table. Among all entries that match the destination address, a router picks the one that has the longest subnet mask.

loopback address
A network address used for testing which causes outgoing data to be processed by the local protocol software without sending packets. IP uses 127.0.0.0 as the loopback prefix.

LSR
(*Loose Source Route*) An IP option that contains a list of router addresses that the datagram must visit in order. Unlike a strict source route, a loose source route allows the datagram to pass through additional routers not on the list. See *SSR.*

MABR
(*Multicast Area Border Router*) The MOSPF term for a multicast router that exchanges routing information with another area.

MAC
(*Media Access Control*) A general reference to the low-level hardware protocols used to access a particular network. The term *MAC address* is often used as a synonym for *physical address.*

mail bridge
Informal term used as a synonym for a *mail gateway.*

mail exchanger
A computer that accepts e-mail; some mail exchangers forward the mail to other computers. DNS has a separate address type for mail exchangers.

mail exploder

Part of an electronic mail system that accepts a piece of mail and a list of addresses as input and sends a copy of the message to each address on the list. Most electronic mail systems incorporate a mail exploder to allow users to define mailing lists locally.

mail gateway

A machine that connects to two or more electronic mail systems (especially dissimilar mail systems on two different networks) and transfers mail messages among them. Mail gateways usually capture an entire mail message, reformat it according to the rules of the destination mail system, and then forward the message.

MAN

(*Metropolitan Area Network*) Any physical network technology that operates at high speeds (usually hundreds of megabits per second through several gigabits per second) over distances sufficient for a metropolitan area. See *LAN* and *WAN*.

Management Information Base

See *MIB*.

martians

Humorous term applied to packets that turn up unexpectedly on the wrong network, often because of incorrect routing tables.

mask

See *subnet mask*.

maximum transfer unit

See *MTU*.

MBONE

(*Multicast BackBONE*). A cooperative agreement among sites to forward multicast datagrams across the Internet by use of IP tunneling.

Mbps

(*Millions of Bits Per Second*) A measure of the rate of data transmission equal to 2^{20} bits per second. Also see *Gbps*, *Kbps*, and *baud*.

MIB

(*Management Information Base*) The set of variables (database) that a system running an SNMP agent maintains. Managers can fetch or store into these variables.

MILNET

(*MILitary NETwork*) Originally part of the ARPANET, MILNET was partitioned in 1984.

MIME

(*Multipurpose Internet Mail Extensions*) A standard used to encode data such as images as printable ASCII text for transmission through e-mail.

mobile IP

A technology developed by the IETF to permit a computer to travel to a new site while retaining its original IP address. The computer contacts a server to obtain a second, temporary address, and then arranges for all datagrams to be forwarded to it.

Mosaic

An early Web browser program.

MOSPF

(*Multicast Open Shortest Path First*) Multicast Extensions to the OSPF routing protocol.

MPLS

(*Multi-Protocol Label Switching*) A technology that uses high speed switching hardware to carry IP datagrams. MPLS is descended from IP switching and label switching.

mrouted

(*Multicast ROUTE Daemon*) A program used with a protocol stack that supports IP multicast to establish multicast routing.

MSL

(*Maximum Segment Lifetime*) The longest time a datagram can survive in the Internet. Protocols use the MSL to guarantee a bound on the time duplicate packets can survive.

MSS

(*Maximum Segment Size*) A term used with TCP. The MSS is the largest amount of data that can be transmitted in one segment. Sender and receiver negotiate maximum segment size at connection startup.

MTU

(*Maximum Transfer Unit* or *Maximum Transmission Unit*) The largest amount of data that can be transferred across a given physical network. The MTU is determined by the network hardware.

multi-homed host

A host using TCP/IP that has connections to two or more physical networks.

multicast

A technique that allows copies of a single packet to be passed to a selected subset of all possible destinations. Some hardware (e.g., Ethernet) supports multicast by allowing a network interface to belong to one or more multicast groups. IP supports an internet multicast facility.

multiplex

To combine data from several sources into a single stream in such a way that it can be separated again later. Multiplexing occurs at many levels. See *demultiplex*.

multiplicative decrease

A technique used by TCP to reduce transmission when congestion occurs. TCP decreases the size of the effective window by half each time a segment is lost.

NACK

(*Negative Acknowledgement*) A response from the recipient of data to the sender of that data to indicate that the transmission was unsuccessful (e.g., that the data was corrupted by transmission errors). Usually, a NACK triggers retransmission of the lost data.

Nagle algorithm

A self-clocking heuristic that clumps outgoing data to improve throughput and avoid silly window syndrome.

NAK

Synonym for *NACK*.

name resolution

The process of mapping a name into a corresponding address. The domain name system provides a mechanism for naming computers in which programs use remote name servers to resolve a machine name into an IP address.

NAP

(*Network Access Point*) One of several physical locations where ISPs interconnect their networks. A NAP also includes a route server that supplies each ISP with reachability information from the routing arbiter system. In addition to NAPs, many ISPs now have private peering arrangements.

NAT

(*Network Address Translation*) A technology that allows hosts with private addresses to communicate with an outside network such as the global Internet.

NBMA

(*Non-Broadcast Multi-Access*). A characteristic of a network that connects multiple computers but does not supply hardware-level broadcast. ATM is the prime example of a NBMA network.

Net 10 address

A general reference to a nonroutable address (i.e., one that is reserved for use in an intranet and not used on the global Internet). The prefix 10.0.0.0 was formerly assigned to ARPANET; it was designated as a nonroutable address when the ARPANET ceased operation.

NetBIOS

(*Network Basic Input Output System*) NetBIOS is the standard interface to networks on IBM PC and compatible personal computers. TCP/IP includes guidelines that describe how to map NetBIOS operations into equivalent TCP/IP operations.

network byte order

The TCP/IP standard for transmission of integers that specifies the most significant byte appears first (big endian). Sending machines are required to translate from the local integer representation to network byte order, and receiving machines are required to translate from network byte order to the local machine representation.

network management

See *MIB* and *SNMP*.

Next Header

A field used in IPv6 to specify the type of the item that follows.

NFS

(*Network File System*) A protocol originally developed by SUN Microsystems, Incorporated that uses IP to allow a set of cooperating computers to access each other's file systems as if they were local.

NIC

(*Network Interface Card*) A hardware device that plugs into the bus on a computer and connects the computer to a network.

NIST

(*National Institute of Standards and Technology*) Formerly, the National Bureau of Standards. NIST is one standards organization within the US that establishes standards for network protocols.

NLA

(*Next Level Aggregation*) In IPv6 addressing, the third most significant set of bits in a unicast address. Also see *TLA*.

NOC

(*Network Operations Center*) Originally, the organization at BBN that monitored and controlled several networks that formed part of the global Internet. Now, used for any organization that manages a network.

nonroutable address

Any address that uses one of the network prefixes which are reserved for use in intranets. Routers in the global Internet will report an error if a datagram containing a nonroutable address accidentally reaches them. See *net-10 address*.

NSAP

(*Network Service Access Point*) An address format that can be encoded in 20 octets. The ATM Forum recommends using NSAP addresses.

NSF

(*National Science Foundation*) A U.S. government agency that funded some of the research and development of the Internet.

NSFNET

(*National Science Foundation NETwork*) Used to describe the Internet backbone in the U.S., which is supported by NSF.

NVT

(*Network Virtual Terminal*) The character-oriented protocol used by TELNET.

OC series standards

A series of standards for the transmission of data over optical fiber. For example, the popular OC3 standard has a bit rate of approximately 155 million bits per second.

octet

An 8-bit unit of data. Although engineers frequently use the term *byte* as a synonym for octet, a byte can be smaller or larger than 8 bits.

one-armed router

An IP router that understands two addressing domains, but only has one physical network connection. One-armed routers are typically used to add security or address translation rather than to forward packets between networks. Also called a *one-armed firewall.*

OSI

(*Open Systems Interconnection*) A reference to protocols developed by ISO as a competitor for TCP/IP. They are not widely deployed or supported.

OSPF

(*Open Shortest Path First*) A link state routing protocol design by the IETF.

OUI

(*Organizationally Unique Identifier*) Part of an address assigned to an organization that manufactures network hardware; the organization assigns a unique address to each device by using its OUI plus a suffix number.

out of band data

Data sent outside the normal delivery path, often used to carry abnormal or error indicators. TCP has an *urgent data* facility for sending out-of-band data.

packet

Used loosely to refer to any small block of data sent across a packet switching network.

packet filter

A mechanism in a router that can be configured to reject some types of packets and admit others. Packet filters are used to create a security firewall.

path MTU

The minimum MTU along a path from the source to destination, which specifies the largest datagram that can be sent along the path without fragmentation. The standard recommends that IP use Path MTU Discovery.

PCM

(*Pulse Code Modulation*) A standard for voice encoding used in digital telephony that produces 8000 8-bit samples per second.

PDN

(*Public Data Network*) A network service offered by a common carrier.

PDU

(*Packet Data Unit*) An ISO term used for either packet or message.

peering arrangement

An cooperative agreement between two ISPs to exchange both reachability information and data packets. In addition to peering at NAPs, large ISPs often have private peering arrangements.

PEM
(*Privacy Enchanced Mail*) A protocol for encrypting e-mail to prevent others from reading messages as they travel across an internet.

perimeter security
A network security mechanism that places a firewall at each connection between a site and outside networks.

physical address
A synonym for *MAC address* or *hardware address*.

PIM-DM
(*Protocol Independent Multicast Dense Mode*) A data-driven multicast routing protocol similar to DVMRP.

PIM-SM
(*Protocol Independent Multicast Sparse Mode*) A demand-driven multicast routing protocol that extends the ideas in CBT.

PING
(*Packet InterNet Groper*) The name of a program used with TCP/IP internets to test reachability of destinations by sending them an ICMP echo request and waiting for a reply. The term is now used like a verb as in, ''please ping host *A* to see if it is alive.''

playback point
The minimum amount of data required in a jitter buffer before playback can begin.

point-to-point network
Any network technology such as a serial line that connects exactly two machines. Point-to-point networks do not require attached computers to have a hardware address.

poison reverse
A heuristic used by distance-vector protocols such as RIP to avoid routing loops. When a route disappears, instead of simply removing the route from its advertisement, a router advertises that the destination is no longer reachable.

POP
(*Post Office Protocol*) The protocol used to access and extract e-mail from a mailbox.

port
See *protocol port*.

positive acknowledgement
Synonym for *acknowledgement*.

POTS
(*Plain Old Telephone Service*) A reference to the standard voice telephone system.

PPP

(*Point to Point Protocol*) A protocol for framing IP when sending across a serial line. Also see *SLIP*.

promiscuous ARP

See *proxy ARP*.

promiscuous mode

A feature of network interface hardware that allows a computer to receive all packets on the network.

protocol

A formal description of message formats and the rules two or more machines must follow to exchange those messages. Protocols can describe low-level details of machine to machine interfaces (e.g., the order in which the bits from a byte are sent across a wire), or high-level exchanges between application programs (e.g., the way in which two programs transfer a file across an internet). Most protocols include both intuitive descriptions of the expected interactions as well as more formal specifications using finite state machine models.

protocol port

The abstraction that TCP/IP transport protocols use to distinguish among multiple destinations within a given host computer. TCP/IP protocols identify ports using small positive integers. Usually, the operating system allows an application program to specify which port it wants to use. Some ports are reserved for standard services (e.g., electronic mail).

provider prefix

An addressing scheme in which an ISP owns a prefix of an address and assigns each customer addresses that begin with the prefix. IPv6 offers provider prefix addressing.

provisioned service

A service that is configured manually.

proxy

Any device or system that acts in place of another (e.g., a proxy Web server acts in place of another Web server).

proxy ARP

The technique in which one machine, usually a router, answers ARP requests intended for another by supplying its own physical address. By pretending to be another machine, the router accepts responsibility for forwarding packets. The purpose of proxy ARP is to allow a site to use a single IP network address with multiple physical networks.

prune

An operation in which a multicast router removes itself from a shared forwarding tree; the opposite of *graft*.

pseudo header

Source and destination IP address information sent in the IP header, but included in a TCP or UDP checksum.

PSN

(*Packet Switching Node*) The formal name of ARPANET packet switches that replaced the original term *IMP*.

PSTN

(*Public Switched Telephone Network*) The standard voice telephone system.

public key encryption

An encryption technique that generates encryption keys in pairs. One of the pair must be kept secret, and one is published.

PUP

(*Parc Universal Packet*) In the internet system developed by Xerox Corporation, a PUP is the fundamental unit of transfer, like an IP datagram is in a TCP/IP internet. The name was derived from the name of the laboratory at which the Xerox internet was developed, the Palo Alto Research Center (PARC).

push

The operation an application performs on a TCP connection to force data to be sent immediately. A bit in the segment header marks pushed data.

PVC

(*Permanent Virtual Circuit*) The type of virtual circuit established by an administrator rather than by software in a computer. Unlike an SVC, a PVC lasts a long time (e.g., weeks or months).

QoS

(*Quality of Service*) Bounds on the loss, delay, jitter, and minimum throughput that a network guarantees to deliver. Some proponents argue that QoS is necessary for real-time traffic.

RA

See *routing arbiter.*

RARP

(*Reverse Address Resolution Protocol*) A protocol that can be used at startup to find an IP address. Although once popular, most computers now use *BOOTP* or *DHCP* instead.

RDP

(*Reliable Datagram Protocol*) A protocol that provides reliable datagram service on top of the standard unreliable datagram service that IP provides. RDP is not among the most widely implemented TCP/IP protocols.

reachability

A network is "reachable" from a given host if a datagram can be sent from the host to a destination on the network. Exterior routing protocols exchange reachability information.

reassembly
The process of collecting all the fragments of an IP datagram and using them to create a copy of the original datagram. The ultimate destination performs reassembly.

RED
(*Random Early Discard*) A technique routers use instead of tail-drop when their queue overflows to improve TCP performance. As the queue fills, the router begins discarding datagrams at random.

redirect
An ICMP message sent from a router to a host on a local network to instruct the host to change a route.

reference model
A description of how layered protocols fit together. TCP/IP uses a 5-layer reference model; earlier protocols used the ISO 7-layer reference model.

regional network
A network that covers a medium-size geographical area such as a few cities or a state.

reliable multicast
A multicast delivery system that guarantees reliable transfer to every member.

reliable transfer
Characteristic of a mechanism that guarantees to deliver data without loss, without corruption, without duplication, and in the same order as it was sent, or to inform the sender that delivery is impossible.

repeater
A hardware device that extends a LAN. A repeater copies electrical signals from one physical network to another. No longer popular.

replay
An error situation in which packets from a previous session are erroneously accepted as part of a later session. Protocols that do not prevent replay are not secure.

reserved address
A synonym for *nonroutable address*.

reset
A segment sent by TCP to report an error.

resolution
See *address resolution*

RFC
(*Request For Comments*) The name of a series of notes that contain surveys, measurements, ideas, techniques, and observations, as well as proposed and accepted TCP/IP protocol standards. RFCs are available on-line.

RIP

(*Routing Information Protocol*) A protocol used to propagate routing information inside an autonomous system. RIP derives from an earlier protocol of the same name developed at Xerox.

RJE

(*Remote Job Entry*) A service that allows submission of a (batch) job from a remote site.

rlogin

(*Remote LOGIN*) The remote login protocol developed for UNIX by Berkeley. Rlogin offers essentially the same service as TELNET.

ROADS

(*Running Out of ADdress Space*) A reference to the possible exhaustion of the IPv4 address space.

round trip time

The total time required to traverse a network from a source computer to a destination and back to the source. TCP uses round trip times to compute a retransmission timer.

route

In general, a route is the path that network traffic takes from its source to its destination. In a TCP/IP internet, each IP datagram is routed independently; routes can change dynamically.

route aggregation

The technique used by routing protocols to combine multiple destinations that have the same next hop into a single entry. A default route provides the highest degree of aggregation.

route server

A server that operates at a NAP and uses BGP to communicate reachability information from the routing arbiter database.

routed

(*Route Daemon*) A program devised for UNIX that implements the RIP protocol. Pronounced "route-d."

router

A special purpose, dedicated computer that attaches to two or more networks and forwards packets from one to the other. In particular, an IP router forwards IP datagrams among the networks to which it connects. A router uses the destination address on a datagram to choose a next-hop to which it forwards the datagram. Researchers originally used the term *gateway*.

router alert

An IP option that causes each intermediate router to examine a datagram even if the datagram is not destined to the router.

router requirements

A document that contains updates to TCP/IP protocols used in routers. See *host requirements.*

routing arbiter

A replicated, authenticated database that contains all possible routes in the Internet. Each ISP that connects to a NAP uses BGP to communicate with a route server to obtain information.

routing loop

An error condition in which a cycle of routers each has the next router in the cycle as the shortest path to a given destination.

RP

(*Rendezvous Point*) The router used as a target for a join request in a demand-driven multicast scheme.

RPB

(*Reverse Path Broadcast*) A synonym for *RPF.*

RPC

(*Remote Procedure Call*) A technology in which a program invokes services across a network by making modified procedure calls. The NFS protocol uses a specific type of RPC.

RPF

(*Reverse Path Forwarding*) A technique used to propagate broadcast packets that ensures there are no routing loops. IP uses reverse path forwarding to propagate subnet broadcast and multicast datagrams.

RPM

(*Reverse Path Multicast*) A general approach to multicasting that uses the TRPB algorithm.

RS

See *route server.*

RS232

A standard by EIA that specifies the electrical characteristics of slow speed interconnections between terminals and computers or between two computers. Although the standard commonly used is RS232C, most people refer to it as RS232.

RST

(*ReSeT*) A common abbreviation for a TCP *reset* segment.

RSVP

(*Resource ReserVation Protocol*) The protocol that allows an endpoint to request a flow with specific QoS; routers along the path to the destination must agree before they approve the request.

RTCP

(*RTP Control Protocol*) The companion protocol to RTP used to control a session.

RTO

(*Round trip Time-Out*) The delay used before retransmission. TCP computes RTO as a function of the current round trip time and variance.

RTP

(*Real-time Transport Protocol*) The primary protocol used to transfer real-time data such as voice and video over IP.

RTT

(*Round Trip Time*) A measure of delay between two hosts. The round trip time consists of the total time taken for a single packet or datagram to leave one machine, reach the other, and return. In most packet switching networks, delays vary as a result of congestion. Thus, a measure of round trip time is an average, which can have high standard deviation.

SA

(*Security Association*) Used with IPsec to denote a binding between a set of security parameters and an identifier carried in a datagram header. A host chooses SA bindings; they are not globally standardized. See *SPI.*

SACK

(*Selective ACKnowledgement*) An acknowledgement mechanism used with sliding window protocols that allows the receiver to acknowledge packets received out of order, but within the current sliding window. Also called extended acknowledgement. Compare to the *cumulative acknowledgement* scheme used by TCP.

SAR

(*Segmentation And Reassembly*) The process of dividing a message into cells, sending them across an ATM network, and reforming the original message. AAL5 performs SAR when sending IP across an ATM network.

segment

The unit of transfer sent from TCP on one machine to TCP on another. Each segment contains part of a stream of bytes being sent between the machines as well as additional fields that identify the current position in the stream and a checksum to ensure validity of received data.

selective acknowledgement

See SACK.

self clocking

Characteristic of any system that operates periodically without requiring an external clock (e.g., uses the arrival of a packet to trigger an action).

self-healing

Characteristic of a mechanism that overcomes failure automatically. A dual FDDI ring is self-healing because it can accommodate failure of a station or a link.

self-identifying frame

Any network frame or packet that includes a field to identify the type of the data being carried. Ethernet uses self-identifying frames, but ATM does not.

server

A running program that supplies service to clients over a network. Examples include providing access to files or to World Wide Web pages.

seven-layer reference model

See *ISO*.

SGMP

(*Simple Gateway Monitoring Protocol*) A predecessor of SNMP.

shared tree

A forwarding scheme used by demand-driven multicast routing protocols. A shared tree is an alternative to a shortest path tree.

shortest path routing

Routing in which datagrams are directed over the shortest path; all routing protocols try to compute shortest paths. Also see *SPF*.

shortest path tree

The multicast forwarding tree that is optimal from a given source to all members of the group. A shortest path trees is an alternative to a shared tree.

signaling

A telephony term that refers to protocols which establish a circuit.

silly window syndrome

A condition that can arise in TCP in which the receiver repeatedly advertises a small window and the sender repeatedly sends a small segment to fill it. The resulting transmission of small segments makes inefficient use of network bandwidth.

SIP

(*Session Initiation Protocol*) A protocol devised by the IEFT for signaling in IP telephony. (Note: SIP was formerly used to refer to *Simple IP*, a protocol that served as the basis for IPv6.)

SIPP

(*SIP Plus*) An extension of *Simple IP* that was proposed for IPv6. See *IPv6*.

site-local address

An address used with IPv6 that has significance only at a single site.

sliding window

Characteristic of protocols that allow a sender to transmit more than one packet of data before receiving an acknowledgement. After receiving an acknowledgement for the first packet sent, the sender "slides" the packet window and sends another. The number of outstanding packets or bytes is known as the window size; increasing the window size improves throughput.

SLIP

(*Serial Line IP*) A framing protocol used to send IP across a serial line. SLIP is popular when sending IP over dialup phone lines. See *PPP*.

slow convergence

A problem in distance-vector protocols in which two or more routers form a routing loop that persists until the routing protocols increment the distance to infinity.

slow-start

A congestion avoidance scheme in TCP in which TCP increases its window size as ACKs arrive. The term is a slight misnomer because slow-start achieves high throughput by using exponential increases.

SMDS

(*Switched Multimegabit Data Service*) A connectionless packet service developed by regional telephone companies.

SMI

(*Structure of Management Information*) Rules that describe the form of MIB variables.

SMTP

(*Simple Mail Transfer Protocol*) The TCP/IP standard protocol for transferring electronic mail messages from one machine to another. SMTP specifies how two mail systems interact and the format of control messages they exchange to transfer mail.

SNA

(*System Network Architecture*) The name applied to an architecture and a class of network products offered by IBM Corporation. SNA does not interoperate with TCP/IP.

SNAP

(*SubNetwork Attachment Point*) An IEEE standard for a small header that is added to data when sending across a network that does not have self-identifying frames. The SNAP header specifies the type of the data.

SNMP

(*Simple Network Management Protocol*) A protocol used to manage devices such as hosts, routers, and printers. A specific version is denoted with a suffix (e.g., SNMPv3). Also see *MIB*.

SOA

(*Start Of Authority*) A keyword used with DNS to denote the beginning of the records for which a particular server is the authority. Other records in the server are reported as non-authoritative answers.

socket API

The set of procedures an application uses to communicate over a TCP/IP network. The name is derived from an abstraction offered by the Unix operating system.

soft state

A technique in which a receiver times out information rather than depending on the sender to maintain it. Soft state works well when the sender and receiver become disconnected.

source quench

A congestion control technique in which a machine experiencing congestion sends a message back to the source of the packets requesting that the source stop transmitting. In a TCP/IP internet, routers send an ICMP source quench message when a datagram overruns the input queue.

source route

A route that is determined by the source. In IP, a source route consists of a list of routers a datagram should visit; the route is specified as an IP option. Source routing is most often used for debugging. See *LSR* and *SSR*.

source tree

A synonym for *shortest path tree*.

SPF

(*Shortest Path First*) A class of routing update protocols that uses Dijkstra's algorithm to compute shortest paths. See *link state routing*.

SPI

(*Security Parameters Index*) The identifier IPsec uses to specify the Security Association that should be used to process a datagram.

split horizon update

A heuristic used by distance-vector protocols such as RIP to avoid routing loops. Routes are not advertised over the interface from which they were learned.

SS7

(*Signaling System 7*) The conventional telephone system standard used for signaling.

SSL

(*Secure Sockets Layer*) A de facto standard for secure communication created by Netscape, Inc. SSL was an Internet Draft, but did not become an RFC.

SSR

(*Strict Source Route*) An IP option that contains a list of router addresses that the datagram must visit in order. See *LSR*.

standard byte order

See *network byte order*.

STD

(*STanDard*) The designation used to classify a particular RFC as describing a standard protocol.

store-and-forward

The paradigm used by IP routers in which an incoming datagram is stored in memory until it can be forwarded on toward its destination.

subnet addressing

An extension of the IP addressing scheme that allows a site to use a single IP network address for multiple physical networks. Outside of the site using subnet addressing, routing continues as usual by dividing the destination address into a network portion and a local portion. Routers and hosts inside a site using subnet addressing interpret the local portion of the address by dividing it into a physical network portion and a host portion.

subnet mask

A bit mask used to select the bits from an IP address that correspond to the subnet. Each mask is 32 bits long, with one bits in the portion that identifies a network and zero bits in the portion that identifies a host.

SubNetwork Attachment Point

See *SNAP*.

supernet addressing

Another name for *CIDR*.

SVC

(*Switched Virtual Circuit*) The type of virtual circuit established dynamically and terminated when no longer needed; usually software in a computer requests an SVC. Unlike a PVC, an SVC can have a short duration.

SWS

See *silly window syndrome*.

SYN

(*SYNchronizing segment*) The first segment sent by the TCP protocol, it is used to synchronize the two ends of a connection in preparation for opening a connection.

T3

The telephony designation for a protocol used over DS3-speed lines. The term is often used (incorrectly) as a synonym for DS3.

tail drop

A policy routers use to manage queue overflow which simply discards all datagrams that arrive after the queue is full. More harmful to TCP throughput than RED.

TCP

(*Transmission Control Protocol*) The TCP/IP standard transport level protocol that provides the reliable, full duplex, stream service on which many application protocols depend. TCP allows a process on one machine to send a stream of data to a process on another. TCP is connection-oriented in the sense that before transmitting data, participants must establish a connection. All data travels in TCP segments, which each travel across the Internet in an IP datagram. The entire protocol suite is often referred to as TCP/IP because TCP and IP are the two fundamental protocols.

TCP/IP Internet Protocol Suite

The official name of the TCP/IP protocols.

TDM

(*Time Division Multiplexing*) A technique used to multiplex multiple signals onto a single hardware transmission channel by allowing each signal to use the channel for a short time before going on to the next one. Also see *FDM*.

TDMA

(*Time Division Multiple Access*) A method of network access in which time is divided into slots and each node on the network is assigned one of the slots. Because all nodes using TDMA must synchronize exactly (even though the network introduces propagation delays between them), TDMA technologies are difficult to design and the equipment is expensive.

TELNET

The TCP/IP standard protocol for remote terminal service. TELNET allows a user at one site to interact with a remote timesharing system at another site as if the user's keyboard and display connected directly to the remote machine.

TFTP

(*Trivial File Transfer Protocol*) The TCP/IP standard protocol for file transfer with minimal capability and minimal overhead. TFTP depends only on the unreliable, connectionless datagram delivery service (UDP), so it is designed for use on a local network.

thicknet

Used to refer to the original thick coaxial cable used with 10Base5 Ethernet. See *thinnet*, *10Base2*, and *10Base-T*.

thinnet

Used to refer to the thinner, more flexible coaxial cable used with 10Base2 Ethernet. See *thicknet*, *10Base5*, and *10Base-T*.

three-way handshake

The 3-segment exchange TCP uses to reliably start or gracefully terminate a connection.

TLA

(*Top Level Aggregation*) In IPv6 addressing, the second most significant set of bits in a unicast address. Also see *NLA*.

TLI

(*Transport Layer Interface*) An alternative to the socket interface defined for System V UNIX.

TLV encoding

Any representation format that encodes each item with three fields: a type, a length, and a value. IP options often use TLV encoding.

tn3270

A version of TELNET for use with IBM 3270 terminals.

token ring

When used in the generic sense, a type of network technology that controls media access by passing a distinguished packet, called a token, from machine to machine. A computer can only transmit a packet when holding the token. When used in a specific sense, it refers to the token ring network hardware produced by IBM.

TOS

(*Type Of Service*) A reference to the original interpretation of the field in an IPv4 header that allows the sender to specify the type of service desired. Now replaced by *DiffServe*.

TP-4

A protocol designed by ISO to be similar to TCP.

traceroute

A program that prints the path to a destination. Traceroute sends a sequence of datagrams with the Time-To-Live set to 1, 2, etc., and uses the ICMP TIME EXCEEDED messages that are returned to determine routers along the path.

traffic class

A reference to a set of services available in the *DiffServe* interpretation.

traffic policing

A reference to mechanisms used with systems that guarantee QoS. Incoming traffic is measured, and any traffic that exceeds the agreed bounds is discarded.

traffic shaping

A reference to mechanisms used with systems that guarantee QoS. Incoming traffic is placed in a buffer and clocked out at a fixed rate.

trailer encapsulation

A nonconventional method of encapsulating IP datagrams for transmission in which the ''header'' information is placed at the end of the packet. Trailers have been used with Ethernet to aid in aligning data on page boundaries. ATM's AAL5 uses trailers.

transceiver

A device that connects a host interface to a local area network (e.g., Ethernet). Ethernet transceivers contain analog electronics that apply signals to the cable and sense collisions.

triggered updates

A heuristic used with distance-vector protocols such as RIP. When a routing table changes, the router sends updates immediately without waiting for the next cycle.

TRPB

(*Truncated Reverse Path Broadcast*) A technique used in data-driven multicasting to forward multicast datagrams. See *broadcast* and *prune*.

TRPF

(*Truncated Reverse Path Forwarding*) A synonym for *TRPB*.

TTL

(*Time To Live*) A technique used in best-effort delivery systems to avoid endlessly looping packets. For example, each IP datagram is assigned an integer time to live when it is created. Each router decrements the time to live field when the datagram arrives, and a router discards any datagram if the time to live counter reaches zero.

tunneling

A technique in which a packet is encapsulated in a high-level protocol and passed across a transport system. The MBONE tunnels each IP multicast datagram inside a conventional IP datagram; a VPN uses tunneling to pass encrypted datagrams between sites. See *IP-in-IP*.

twisted pair Ethernet

The 10Base-T Ethernet wiring scheme that uses twisted pair wires from each computer to a hub. See *thicknet* and *thinnet*.

type of service routing

A routing scheme in which the choice of path depends on the characteristics of the underlying network technology as well as the shortest path to the destination.

UART

(*Universal Asynchronous Receiver and Transmitter*) An electronic device consisting of a single chip that can send or receive characters on asynchronous serial communication lines that use RS232. UARTs are flexible because they have control lines that allow the designer to select parameters like transmission speed, parity, number of stop bits, and modem control. UARTs appear in terminals, modems, and on the I/O boards in computers that connect the computer to terminal(s).

UCBCAST

See *Berkeley broadcast*.

UDP

(*User Datagram Protocol*) The protocol that allows an application program on one machine to send a datagram to an application program on another. UDP uses the Internet Protocol (IP) to deliver datagrams. Conceptually, the important difference between UDP datagrams and IP datagrams is that UDP includes a protocol port number, allowing the sender to distinguish among multiple application programs on a given remote machine.

unicast

A method of addressing and routing in which a packet is delivered to a single destination. Most IP datagrams are sent via unicast. See *multicast*.

universal time

The international standard time reference that was formerly called Greenwich Mean Time. It is also called universal coordinated time.

unnumbered network

A technique for conserving IP network prefixes that leaves a point to point connection between two routers unnumbered.

unreliable delivery

Characteristic of a mechanism that does not guarantee to deliver data without loss, corruption, duplication, or in the same order as it was sent. IP is unreliable.

urgent data

The method used in TCP to send data out of band. A receiver processes urgent data immediately upon receipt.

URI

(*Uniform Resource Identifier*) A generic term used to refer to a URN or a URL.

URL

(*Uniform Resource Locator*) A string that gives the location of a piece of information. The string begins with a protocol type (e.g., FTP) followed by the identification of specific information (e.g., the domain name of a server and the path name to a file on that server).

URN

(*Uniform Resource Name*) A string that gives the location of a piece of information. Unlike a URL, a URN is guaranteed to persist over long periods of time.

UUCP

(*Unix to Unix Copy Program*) An application program developed in the mid 1970s for version 7 UNIX that allows one UNIX timesharing system to copy files to or from another UNIX timesharing system over a single (usually dialup) link. Because UUCP is the basis for electronic mail transfer in UNIX, the term is often used loosely to refer to UNIX mail transfer.

variable-length subnetting

A subnet address assignment scheme in which each physical net in an organization can have a different mask. The alternative is *fixed-length subnetting*.

vBNS

(*very high speed Backbone Network Service*) The 155 Mbps backbone network that was deployed in 1995 and is now used for networking research.

VC

(*Virtual Circuit*) A path through a network from one application to another that is used to send data. The VC, established either by protocol software or manually, provides the illusion of a "connection". Although the concept is the same, ATM expands the term to *Virtual Channel*.

vector-distance

Now called *distance-vector*.

very high speed Backbone Network Service

See vBNS.

virtual circuit

The basic abstraction provided by a connection-oriented protocol like TCP. Once a virtual circuit has been created, it stays in effect until explicitly shut down.

VLSM

(*Variable Length Subnet Mask*) A subnet mask used with variable length subnetting.

VPI/VCI

(*Virtual Path Identifier* plus *Virtual Circuit Identifier*) A connection identifier used by ATM; each connection a host opens is assigned a unique VPI/VCI.

VPN

(*Virtual Private Network*) A technology that connects two or more separate sites over the Internet, but allows them to function as if they were a single, private network. VPN software guarantees that although packets travel across the Internet, the contents remains private.

WAN

(*Wide Area Network*) Any physical network technology that spans large geographic distances. Also called long-haul networks, WANs have significantly higher delays and higher costs than networks that operate over shorter distances. See *LAN* and *MAN*.

well-known port

Any of a set of protocol port numbers preassigned for specific uses by transport level protocols (i.e., TCP and UDP). Each server listens at a well-known port, so clients can locate it.

window

See *sliding window*.

window advertisement

A value used by TCP to allow a receiver to tell a sender the size of an available buffer.

Windows Sockets Interface

A variant of the socket API developed by Microsoft. Often called *WINSOCK*.

working group

A group of people in the IETF working on a particular protocol or design issue.

World Wide Web

The large hypermedia service available on the Internet that allows a user to browse information.

WWW

See *World Wide Web*.

X

See *X-Window System*.

X.25

An older protocol standardized by the ITU which was popular in Europe before TCP/IP.

X25NET

(*X.25 NETwork*) A service offered by CSNET that passed IP traffic between a subscriber site and the Internet using X.25.

X.400

The ITU protocol for electronic mail.

XDR

(*eXternal Data Representation*) The standard for a machine-independent data representation. To use XDR, a sender translates from the local machine representation to the standard external representation and a receiver translates from the external representation to the local machine representation.

X-Window System

A software system developed at MIT for presenting and managing output on bit-mapped displays. Each window consists of a rectangular region of the display that contains textual or graphical output from one remote program. A special program called a window manager allows the user to create, move, overlap, and destroy windows.

zero window

See *closed window*.

zone of authority

Term used in the domain name system to refer to the group of names for which a given name server is an authority. Each zone must be supplied by two name servers that have no common point of failure.

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